

TRANSCRIBE PROJECT DOCUMENTS

Backend Schema

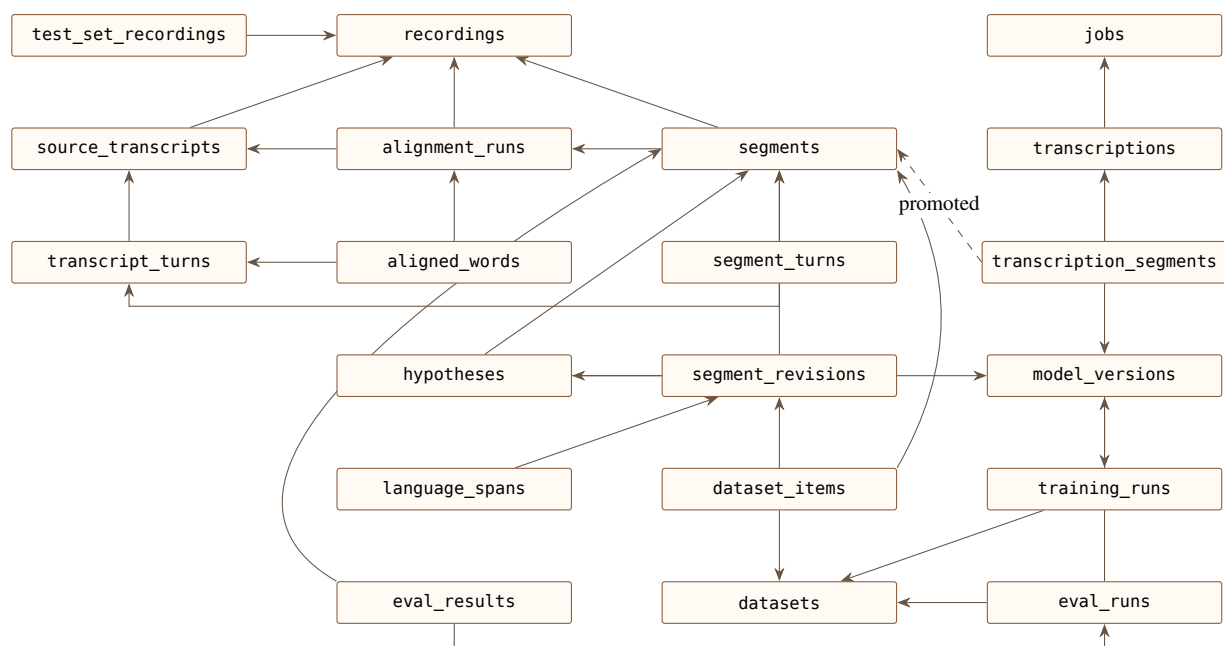
Document 05

Status	Draft v1
Database	SQLite 3 (WAL mode) via SQLAlchemy 2 + Alembic. Portable to PostgreSQL without schema changes (no SQLite-only types; JSON is stored as TEXT with CHECK json_valid, which becomes JSONB on Postgres)
Related	02 TRD , 04 App Flow

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1 Entity overview

**Figure 1:** Entity overview. Arrows point from the referencing table to the referenced table.

Conventions:

- IDs are ULIDs stored as TEXT (sortable by creation time and safe to generate in either process).
- Timestamps are ISO-8601 UTC strings in TEXT (created_at, updated_at).
- Audio times are REAL seconds relative to the recording start.
- Enumerations are TEXT columns with CHECK constraints; in SQLAlchemy they map to Enum(..., native_enum=False).

2 DDL

```
PRAGMA foreign_keys = ON;
PRAGMA journal_mode = WAL;

-- ----- Sources -----
CREATE TABLE recordings (
  id          TEXT PRIMARY KEY,
  title       TEXT NOT NULL,
  original_path TEXT NOT NULL,          -- data/store/originals/<sha256>.<ext>
  normalized_path TEXT,                -- data/store/audio16k/<sha256>.wav
  sha256      TEXT NOT NULL UNIQUE,
  duration_s  REAL NOT NULL,
  domain      TEXT NOT NULL DEFAULT 'interview'
              CHECK (domain IN ('interview', 'tv', 'radio', 'church', 'meeting', 'other')),
  consent_scope TEXT NOT NULL
              CHECK (consent_scope IN ('training', 'eval_only', 'transcription_only')),
  recorded_on  TEXT,
  region       TEXT,                  -- free text, optional; never ethnicity
  status       TEXT NOT NULL DEFAULT 'imported'
              CHECK (status IN ('imported', 'aligning', 'ready_for_review',
                                'alignment_suspect', 'reviewed', 'archived')),
  notes        TEXT,
  created_at   TEXT NOT NULL,
  updated_at   TEXT NOT NULL
);

CREATE TABLE source_transcripts (
  id          TEXT PRIMARY KEY,
  recording_id TEXT NOT NULL REFERENCES recordings(id) ON DELETE CASCADE,
  path        TEXT NOT NULL,
  format      TEXT NOT NULL CHECK (format IN ('docx', 'pdf', 'txt')),
  sha256      TEXT NOT NULL,
  parser      TEXT NOT NULL,          -- 'python-docx' | 'pypdf'
  parser_version TEXT NOT NULL,
  italic_means TEXT CHECK (italic_means IN ('sw', 'en', 'none')) DEFAULT 'sw',
  created_at  TEXT NOT NULL,
  UNIQUE (recording_id, sha256)
);

CREATE TABLE transcript_turns (
  id          TEXT PRIMARY KEY,
  source_transcript_id TEXT NOT NULL REFERENCES source_transcripts(id) ON DELETE CASCADE,
  ordinal     INTEGER NOT NULL,
  speaker_label TEXT,                -- 'I', 'R', 'P1' as written
  speaker_role TEXT CHECK (speaker_role IN ('interviewer', 'respondent', 'host',
                                             'preacher', 'audience', 'other')),
  text_raw    TEXT NOT NULL,          -- exactly as in the source (label source)
  text_align  TEXT NOT NULL,          -- normalised text used by the aligner
  word_map    TEXT NOT NULL CHECK (json_valid(word_map)),
              -- [[align_word_idx, raw_char_start, raw_char_end], ...]
  spans       TEXT NOT NULL DEFAULT '[]' CHECK (json_valid(spans)),
```

```

-- language hints from formatting: [{start,end,lang}]
annotations    TEXT NOT NULL DEFAULT '[]' CHECK (json_valid(annotations)),
-- [{"marker":"[unclear]","char":123}]
UNIQUE (source_transcript_id, ordinal)
);

-- ----- Alignment -----
CREATE TABLE alignment_runs (
  id            TEXT PRIMARY KEY,
  recording_id  TEXT NOT NULL REFERENCES recordings(id) ON DELETE CASCADE,
  source_transcript_id TEXT NOT NULL REFERENCES source_transcripts(id),
  aligner       TEXT NOT NULL,           -- 'ctc-forced-aligner'
  aligner_version TEXT NOT NULL,
  params        TEXT NOT NULL CHECK (json_valid(params)),
  mean_confidence REAL,
  status        TEXT NOT NULL CHECK (status IN ('running','done','failed')),
  created_at    TEXT NOT NULL,
  finished_at   TEXT
);

CREATE TABLE aligned_words (
  alignment_run_id TEXT NOT NULL REFERENCES alignment_runs(id) ON DELETE CASCADE,
  turn_id          TEXT NOT NULL REFERENCES transcript_turns(id),
  word_idx         INTEGER NOT NULL,      -- index within turn's text_align
  word             TEXT NOT NULL,
  start_s          REAL NOT NULL,
  end_s            REAL NOT NULL,
  score            REAL NOT NULL,        -- 0..1
  PRIMARY KEY (alignment_run_id, turn_id, word_idx)
);

-- ----- Segments & review -----
CREATE TABLE segments (
  id            TEXT PRIMARY KEY,
  recording_id  TEXT NOT NULL REFERENCES recordings(id) ON DELETE CASCADE,
  alignment_run_id TEXT REFERENCES alignment_runs(id),
  origin        TEXT NOT NULL CHECK (origin IN ('alignment','vad','inference_correction')),
  ordinal       INTEGER NOT NULL,
  start_s       REAL NOT NULL,           -- mirrors current revision
  end_s         REAL NOT NULL,
  speaker_label TEXT,
  status        TEXT NOT NULL DEFAULT 'candidate'
  CHECK (status IN ('candidate','in_review','draft','approved',
    'approved_eval_only','rejected')),
  current_revision_id TEXT,              -- FK added below (circular)
  alignment_conf REAL,
  auto_flags     TEXT NOT NULL DEFAULT '[]' CHECK (json_valid(auto_flags)),
  -- ['low_alignment','unclear_marker','overlap_marker',...]
  priority       REAL NOT NULL DEFAULT 0,
  audio_cache_path TEXT,                 -- rendered slice; invalidated on bound change
  created_at     TEXT NOT NULL,
  updated_at     TEXT NOT NULL,
  CHECK (end_s > start_s),
  UNIQUE (recording_id, ordinal)
);

CREATE INDEX ix_segments_queue ON segments (recording_id, status, priority DESC);

CREATE TABLE segment_turns (
  segment_id      TEXT NOT NULL REFERENCES segments(id) ON DELETE CASCADE,
  turn_id         TEXT NOT NULL REFERENCES transcript_turns(id),

```

```

PRIMARY KEY (segment_id, turn_id)
);

CREATE TABLE segment_revisions (
    id TEXT PRIMARY KEY,
    segment_id TEXT NOT NULL REFERENCES segments(id) ON DELETE CASCADE,
    revision_no INTEGER NOT NULL,
    parent_revision_id TEXT REFERENCES segment_revisions(id),
    text TEXT NOT NULL,
    start_s REAL NOT NULL,
    end_s REAL NOT NULL,
    speaker_label TEXT,
    flags TEXT NOT NULL DEFAULT '{}' CHECK (json_valid(flags)),
    -- {"unclear":false,"overlap":false,"noisy":false,
    -- "music":false,"non_verbatim":false}
    status TEXT NOT NULL,
    source TEXT NOT NULL CHECK (source IN ('aligned_transcript','vad_empty',
    'model_hypothesis','human')),
    hypothesis_id TEXT REFERENCES hypotheses(id),
    author TEXT NOT NULL,
    created_at TEXT NOT NULL,
    UNIQUE (segment_id, revision_no)
);

CREATE TABLE language_spans (
    revision_id TEXT NOT NULL REFERENCES segment_revisions(id) ON DELETE CASCADE,
    char_start INTEGER NOT NULL,
    char_end INTEGER NOT NULL,
    lang TEXT NOT NULL CHECK (lang IN ('en','sw','mixed','luo','other','unclear')),
    PRIMARY KEY (revision_id, char_start),
    CHECK (char_end > char_start)
);

CREATE TABLE hypotheses (
    id TEXT PRIMARY KEY,
    segment_id TEXT NOT NULL REFERENCES segments(id) ON DELETE CASCADE,
    model_version_id TEXT NOT NULL REFERENCES model_versions(id),
    text TEXT NOT NULL,
    words TEXT CHECK (json_valid(words)),
    avg_logprob REAL,
    no_speech_prob REAL,
    wer_vs_current REAL,
    created_at TEXT NOT NULL
);

-- ----- Datasets, training, models -----
CREATE TABLE datasets (
    id TEXT PRIMARY KEY,
    name TEXT NOT NULL UNIQUE,
    filter_spec TEXT NOT NULL CHECK (json_valid(filter_spec)),
    split_spec TEXT NOT NULL CHECK (json_valid(split_spec)),
    test_set_name TEXT NOT NULL,
    normalizer_version TEXT NOT NULL,
    sha256 TEXT,
    frozen INTEGER NOT NULL DEFAULT 0 CHECK (frozen IN (0,1)),
    export_path TEXT,
    stats TEXT CHECK (json_valid(stats)),
    created_at TEXT NOT NULL
);

```

```

CREATE TABLE dataset_items (
  dataset_id      TEXT NOT NULL REFERENCES datasets(id) ON DELETE CASCADE,
  segment_id      TEXT NOT NULL REFERENCES segments(id),
  revision_id     TEXT NOT NULL REFERENCES segment_revisions(id), -- pinned text
  split           TEXT NOT NULL CHECK (split IN ('train','dev','test')),
  PRIMARY KEY (dataset_id, segment_id)
);

CREATE INDEX ix_dataset_items_split ON dataset_items (dataset_id, split);

CREATE TABLE test_set_recordings (          -- frozen test membership, by recording
  test_set_name   TEXT NOT NULL,
  recording_id    TEXT NOT NULL REFERENCES recordings(id),
  PRIMARY KEY (test_set_name, recording_id)
);

CREATE TABLE training_runs (
  id              TEXT PRIMARY KEY,
  dataset_id      TEXT NOT NULL REFERENCES datasets(id),
  base_model      TEXT NOT NULL,              -- 'openai/whisper-small'
  platform        TEXT NOT NULL CHECK (platform IN ('kaggle','local','other')),
  config          TEXT NOT NULL CHECK (json_valid(config)),
  code_commit     TEXT,                      -- git sha of transcribe package used
  notebook_ref    TEXT,                      -- kaggle notebook slug + version
  dev_metrics     TEXT CHECK (json_valid(dev_metrics)),
  started_at      TEXT,
  finished_at     TEXT,
  status          TEXT NOT NULL CHECK (status IN ('registered','failed'))
);

CREATE TABLE model_versions (
  id              TEXT PRIMARY KEY,
  name            TEXT NOT NULL UNIQUE,      -- 'whisper-small-kct-v1'
  family          TEXT NOT NULL CHECK (family IN ('whisper','w2vbert_ctc','legacy_ctc')),
  base_model      TEXT NOT NULL,
  training_run_id TEXT REFERENCES training_runs(id), -- NULL for zero-shot baselines
  hf_path         TEXT NOT NULL,             -- data/store/models/<name>/hf
  ct2_path        TEXT,                     -- data/store/models/<name>/ct2-int8
  decode_params   TEXT NOT NULL CHECK (json_valid(decode_params)),
  status          TEXT NOT NULL DEFAULT 'candidate'
                      CHECK (status IN ('converting','candidate','production','retired','broken')),
  promoted_at     TEXT,
  promotion_note  TEXT,
  created_at      TEXT NOT NULL
);

CREATE UNIQUE INDEX ux_one_production ON model_versions (status) WHERE status = 'production';

CREATE TABLE eval_runs (
  id              TEXT PRIMARY KEY,
  model_version_id TEXT NOT NULL REFERENCES model_versions(id),
  dataset_id      TEXT NOT NULL REFERENCES datasets(id),
  split           TEXT NOT NULL CHECK (split IN ('dev','test','external')),
  normalizer_version TEXT NOT NULL,
  metrics         TEXT CHECK (json_valid(metrics)),
                      -- {"wer":...,"cer":...,"wer_en":...,"wer_sw":...,"wer_mixed":...,"switch_point_er":...,"halluc_per_min":...,"rtf":...,"by_domain":{"..},"by_flag":{"..}}
  status          TEXT NOT NULL CHECK (status IN ('queued','running','done','failed')),
  created_at      TEXT NOT NULL
);

```

```

CREATE TABLE eval_results (
  eval_run_id    TEXT NOT NULL REFERENCES eval_runs(id) ON DELETE CASCADE,
  segment_id    TEXT NOT NULL REFERENCES segments(id),
  reference      TEXT NOT NULL,
  hypothesis     TEXT NOT NULL,
  wer            REAL NOT NULL,
  cer           REAL NOT NULL,
  n_ref_words   INTEGER NOT NULL,
  n_switch_points INTEGER NOT NULL,
  n_switch_errors INTEGER NOT NULL,
  PRIMARY KEY (eval_run_id, segment_id)
);

-- ----- Jobs & transcription -----
CREATE TABLE jobs (
  id            TEXT PRIMARY KEY,
  kind          TEXT NOT NULL CHECK (kind IN ('parse_transcript','align','vad_segment',
      'hypothesize','build_dataset','export_dataset','register_model',
      'convert_model','evaluate','transcribe','export_transcription')),
  payload       TEXT NOT NULL CHECK (json_valid(payload)),
  status        TEXT NOT NULL DEFAULT 'queued'
      CHECK (status IN ('queued','running','done','failed','cancelled')),
  progress      REAL NOT NULL DEFAULT 0, -- 0..1
  attempts      INTEGER NOT NULL DEFAULT 0,
  worker_id     TEXT,
  heartbeat_at  TEXT,
  error         TEXT,
  result        TEXT CHECK (json_valid(result)),
  created_at    TEXT NOT NULL,
  started_at    TEXT,
  finished_at   TEXT
);
CREATE INDEX ix_jobs_queue ON jobs (status, created_at);

CREATE TABLE transcriptions (
  id            TEXT PRIMARY KEY,
  job_id        TEXT NOT NULL REFERENCES jobs(id),
  title         TEXT NOT NULL,
  input_path    TEXT NOT NULL,
  sha256        TEXT NOT NULL,
  duration_s    REAL,
  domain        TEXT NOT NULL DEFAULT 'other',
  consent_to_train INTEGER NOT NULL DEFAULT 0 CHECK (consent_to_train IN (0,1)),
  model_version_id TEXT NOT NULL REFERENCES model_versions(id),
  rtf           REAL,
  expires_at    TEXT, -- retention; NULL once promoted
  created_at    TEXT NOT NULL
);

CREATE TABLE transcription_segments (
  id            TEXT PRIMARY KEY,
  transcription_id TEXT NOT NULL REFERENCES transcriptions(id) ON DELETE CASCADE,
  ordinal       INTEGER NOT NULL,
  start_s       REAL NOT NULL,
  end_s         REAL NOT NULL,
  text          TEXT NOT NULL, -- raw model output (immutable)
  words         TEXT CHECK (json_valid(words)),
  avg_logprob   REAL,
  no_speech_prob REAL,
  edited_text    TEXT, -- user edit; exports prefer this

```

```

edited_at      TEXT,
promoted_segment_id TEXT REFERENCES segments(id),
UNIQUE (transcription_id, ordinal)
);

```

The circular foreign key `segments.current_revision_id → segment_revisions.id` is created by Alembic with `use_alter=True`. SQLite does not enforce ALTER-added foreign keys, so the invariant is checked in the service layer and by a test.

3 Key invariants (enforced in the service layer and tested)

1. `segment_revisions` is append-only: no UPDATE or DELETE except by cascade when a recording is deleted.
2. `segments.{start_s,end_s,status,speaker_label}` always equal those of `current_revision_id`.
3. A hypotheses row never changes `segment_revisions`. Only an explicit reviewer action creates a revision with `source='model_hypothesis'`.
4. `datasets.frozen = 1` means the dataset has no inserts, updates or deletes on `dataset_items`, and `sha256` is set.
5. The train/dev splits contain only recordings with `consent_scope='training'`. The test split may contain training or eval_only.
6. No `recording_id` appears in more than one split in a dataset, and no recording in `test_set_recordings` appears in train or dev of any dataset.
7. Items with status `approved_eval_only`, or with flags `unclear`, `non_verbatim` or `overlap`, never go to train (they may go to test, tagged by flag).
8. At most one `model_versions` row has `status='production'` (partial unique index).

4 File store layout

```

data/
  transcribe.db
  store/
    originals/<sha256>.<ext>          # byte-identical uploads (never modified)
    transcripts/<sha256>.<docx|pdf>
    audio16k/<sha256>.wav            # normalised full recordings
    slices/<segment_id>_<rev>.wav     # cache; safe to delete
    models/<name>/{hf/, ct2-int8/, run_manifest.json}
    uploads/<transcription_id>.<ext>  # retention-managed
  exports/
    <dataset_name>/
      dataset_card.md
      metadata.jsonl                # see §5
      train/<segment_id>.wav
      dev/<segment_id>.wav
      test/<segment_id>.wav

```

The existing `data/projects/<name>/{transcript.json, audio.json, chunks/}` manifests are migrated once by `python -m transcribe.db.migrate_json` (see the Implementation Plan, Phase 0).

5 Export format (Kaggle / Hugging Face)

`metadata.jsonl` has one line per item and follows the `audiofolder` convention, so the export loads with `load_dataset("audiofolder", data_dir=...)`:

```
{
  "file_name": "train/01J8Z...wav",
  "split": "train",
  "text": "Sasa mimi nilikuwa nimeapply hiyo  

  ↳ job lakini they never called back",
  "segment_id": "01J8Z...",
  "revision_id": "01J90...",
  "recording_group": "r_7f3a",
  "domain": "interview",
  "duration": 11.55,
  "speaker_role": "respondent",
  "flags": [],
  "lang_spans": [[0, 26, "sw"], [27, 36, "mixed"], [37, 40, "sw"], [41, 70, "en"]]
}
```

- recording_group is a salted hash of recording_id, so no titles or file names leave the machine.
- dataset_card.md records the dataset name, sha256, normaliser version, per-split hours, filter spec, and the consent statement.

The Kaggle notebook writes run_manifest.json:

```
{
  "dataset_name": "kct-ds-v3",
  "dataset_sha256": "...",
  "base_model": "openai/whisper-small",
  "code_commit": "...",
  "config": {...},
  "best_step": 3500,
  "dev_wer": 0.231,
  "started_at": "...",
  "finished_at": "...",
  "kaggle_notebook": "user/finetune-whisper@v7"
}
```

6 Pydantic API schemas (core)

```
class RevisionIn(BaseModel):
    base_revision_id: str
    text: str
    start_s: float
    end_s: float
    speaker_label: str | None = None
    flags: dict[Literal["unclear", "overlap", "noisy", "music", "non_verbatim"], bool] = {}
    spans: list[tuple[int, int, Literal["en", "sw", "mixed", "luo", "other", "unclear"]]] = []
    status: Literal["draft", "approved", "approved_eval_only", "rejected"]

class SegmentOut(BaseModel):
    id: str; recording_id: str; ordinal: int
    start_s: float; end_s: float; status: str; priority: float
    auto_flags: list[str]; alignment_conf: float | None
    current_revision: RevisionOut
    hypotheses: list[HypothesisOut]
    audio_url: str

class DatasetIn(BaseModel):
    name: str
    filter_spec: FilterSpec      # recordings, domains, statuses, exclude_flags, roles, min/max
    ↳ duration
    split_spec: SplitSpec      # dev_fraction, seed, group_by: Literal["recording",
    ↳ speaker_cluster"]
    test_set_name: str = "test_v1"
```

7 Migration from the current JSON manifests

Current (audio.json / transcript.json)	New
recording_id, source_audio	recordings (sha256 computed, original copied to store)
transcript_turns[]	source_transcripts + transcript_turns (text → text_raw, spans kept)
segments[] with status=candidate and empty transcript	Dropped. They are VAD candidates that were never reviewed, and re-running alignment gives better segments

Current (audio.json / transcript.json)	New
segments[] with status=approved asr_hypothesis, wer	segments (origin=vad) + revision r1 (source=human, status approved) Ignored (no model existed)
